Fine-tuning ResNet18 for Pet Breed Classification Using the Oxford-IIIT Pet Dataset

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Abstract

Training deep convolutional neural networks from scratch requires large amounts of data as well as significant computational resources, which is why transfer learning is a good solution. This project explores transfer learning for the task of classifying 37 cat and dog breeds using the Oxford-IIIT Pet Dataset by fine-tuning ResNet18, which is pre-trained on ImageNet. Two fine-tuning stategies are explored: simultaneously unfreezing l layers from the start of training, and gradual unfreezing where the layers are instead progressively unfrozen during training. Linear probing, where all the convolutional layers are frozen and only the fully connected layer is trained, achieved 88.88% test accuracy, which demonstrates the strength of pre-trained ImageNet features. Fine-tuning with $l = 3$ unfrozen layers achieved the highest performance of 89.04% and gradual unfreezing achieved 90.24%, both outperforming the linear probe baseline. The project also investigates how performance is affected by limited training data using 100%, 10% and 1% of the training set. The results show that for limited data, the gradual unfreezing strategy is more robust, outperforming the $l = 3$ strategy at both 10% (71.21% vs. 65.25%) and 1% (28.92% vs. 20.63%) of the data. Data augmentation and L2 regularization were found to provide small improvements particularly for the limited data. Finally, we also simulate imbalanced datasets by reducing the training images for cat breeds to 20% of the original amount. This reduced the model's performance for classifying cat breeds (cat F1: 0.74) compared to dog breeds (dog F1: 0.90). Weighted cross-entropy loss was shown to partially mitigate this imbalance, improving cat F1-score to 0.76 while also improving accuracy.

1 Introduction

Image classification is one of the important tasks in computer vision and is used in several areas such as facial recognition and medical imaging. Deep convolutional neural networks, known as CNNs, are able to achieve extremely high performance in these tasks, however training reliable models from scratch require large datasets, long training times and considerable computational resources. This means that training such models can be a challenge if you have limited data or hardware available. One solution to these problems is transfer learning.

Transfer learning uses networks that are pre-trained on large datasets such as ImageNet (1) but adapts them to new tasks, instead of training a model from scratch. This is possible since pre-trained models already learn general features of images such as edges and shapes, which means they are able to achieve strong performance even with smaller datasets and shorter training times.

This project explores the possibilities transfer learning by fine-tuning the ResNet18 convolutional neural network (CNN) with the ultimate goal of classifying cat and dog breeds using the Oxford-IIIT Pet Dataset (2). There are several methods of fine-tuning and regularization when doing transfer learning. This project investigates fine-tuning by unfreezing l-layers as well as gradual unfreezing. The project also examines how reduced data affects the accuracy of the model and how regularization methods such as L2 regularization and data augmentation can improve performance and mitigate overfitting. Finally, our project also creates an imbalanced dataset of only 20% of the cat classes to simulate a real-world scenario of having limited data, and weighted cross-entropy loss is evaluated as a method for handling class imbalance.

2 Related Work

Transfer learning has become a standard method in computer vision tasks, allowing models pre-trained on large datasets such as ImageNet (1) to be adapted and modified for different tasks with limited data. Yosinki et al. (3) researched transferability of different features across different layers in deep networks and showing that early/lower layers learn more general features such as Gabor filters and color blobs while the final/higher layers learn more task-specific features.

Deep Residual Network (ResNets) were originally introduced by He et al. (4) as a solution to the "degradation problem" in deep learning where stacking too many layers to a neural network caused accuracy to degrade. ResNet solved this problem by introducing skip connections that allow gradients to flow directly through the network which enabled training of significantly deeper architectures. ResNet18, used in this project, is the smallest variant and is usually pre-trained on the ImageNet dataset and has been trained on over a million images, and is commonly used for transfer learning tasks thanks to its fast performance and computational efficiency.

Previous work by Hu and You (5) showed that transfer learning using ResNet18 achieved a strong performance on classifying animal types. Hu and You managed to obtain 92% classification accuracy for all eight animals in the dataset by only fine-tuning the fully connected layer.

The Oxford-IIIT Pet Dataset used in our project was created and introduced by Parkhi et al. (2) with 37 cat and dog breed categories with roughly 200 images per class.

3 Data

The Oxford-IIIT Pet-dataset (2) consists of 37 classes with 200 images in each class. The classes represent different breeds of cats and dogs, with large variations in pose, lighting, background, and image quality. These variations make the dataset suitable for evaluating image classification models in realistic conditions.

This project utilized PyTorch methods to load and handle the data from the Oxford-IIIT Pet Dataset in a way that was appropriate for training neural networks. The dataset used was divided into training, validation and test subsets to allow a proper evaluation of performance and to prevent overfitting. Before training, the images were transformed with the help of PyTorch. For the baseline training set (non-augmented), the images were first resized to 256 pixels to ensure a consistent scale, and then center cropped to 224×224 pixels to match ResNet18's expected input dimensions while preserving the central content of each image. This same transformation was done for the validation set as well.

Our project also explored how different amounts of training data affects the performance of transfer learning. To simulate this, smaller subsets were created of the dataset, namely 100%, 10% and 1% of the original training set. In addition to this, imbalanced class distributions were also created artificially to simulate that some classes in real-world datasets are underrepresented. As mentioned earlier, we created this imbalance by only using 20% of the cat classes.

To reduce overfitting and improve the models performance, two different training transforms were compared. Our baseline transform resized images to 256 pixels before applying a fixed center crop to 224×224 pixels. The augmented transform instead used PyTorch's RandomResizedCrop() function, which randomly selects a region of varying size and aspect ratio before resizing to 224×224, meaning the model sees a different crop of each image every epoch. Additionally, the augmented transform used RandomHorizontalFlip() and RandomRotation(15) which randomly flips the image horizontally and applies random rotation up to 15 degrees. These augmentations were applied during training, meaning the model saw a different transformed version of each image every epoch without needing to increase the size of the stored dataset.

4 Methods

4.1 Fine-tuning

In transfer learning, fine-tuning is the act of unfreezing the last couple of layers in a pre-trained network. This allows the network to adjust the weights to fit the new specific classification problem. The number of layers from the last l you decide to unfreeze can impact the performance of the classifier. Too few layers and the network may not be given enough space to properly adjust to the new classification problem. Too many and you risk removing the valuable weights that are able to extract general semantic information from the images. In this project, several models are trained with varying amounts of unfrozen layers to account for this.

Another method also explored in this project is gradual unfreezing. This method introduces a schedule during training, unfreezing layers one at a time once training reaches a certain number of epochs. This allows the later parts of the network responsible for the specific classification task to be completely re-trained towards the new dataset while allowing the earlier layers to keep part of their well initialized weights but also nudge them slightly.

4.2 Fine-tuning with limited data

The main problem with neural networks is that they require a lot of data to reach a good performance. This obviously extends to transfer learning. The project explored how different amounts of data (100%, 10%, 1%) affected performance on our Pet-dataset classifier. This was done by creating a stratified train dataset where importantly the relative percent of each class remained the same.

One of the main ways to still be able to train despite limited data is through using regularization techniques. One of the regularization techniques tested in this project was image augmentation. Data/image augmentation is a powerful, well researched and intuitive way of regularizing image classifiers despite limited data. By applying small augmentations to the images during training, the network is discouraged to simply learn the train set outright, and is instead forced to learn some semantic information about the data. In this project, we applied random resized crops, random horizontal flips, random rotations of up to 15 degrees. Another widely common way of regularization is L2 regularization. L2 regularization penalizes too large weights in the network. This prevents some paths from becoming too dominant and increases overall stability. Both of these regularization techniques were tested on 100%, 10% and 1% of the data.

4.3 Fine-tuning with imbalanced classes

A common problem with real world datasets is that the classes are often imbalanced. This project simulated this by only using 20% of the cats from the dataset and doing some per class evaluations on the resulting networks. There are however methods for adjusting to such datasets. One explored in this project is weighted cross-entropy that assigns a larger penalty for miss-classifying rarer classes.

5 Experiments

5.1 Fine-tuning

As a sanity check, we first replaced the final layer and trained only this layer on a cats vs dogs binary classification task, while keeping the rest of the network frozen. This achieved an accuracy of 98.45%, showing that the pre-trained features are highly effective. We then performed linear probing on the full 37-class problem by training only a new final layer while freezing all convolutional layers. This resulted in an accuracy of 88.88%, confirming strong performance of the pre-trained representations even without fine-tuning.

When first doing the fine-tuning, several models were made, all with different amounts of layers unfrozen from the initial ResNet18 network. This project unfroze three layers before experiencing diminishing returns.

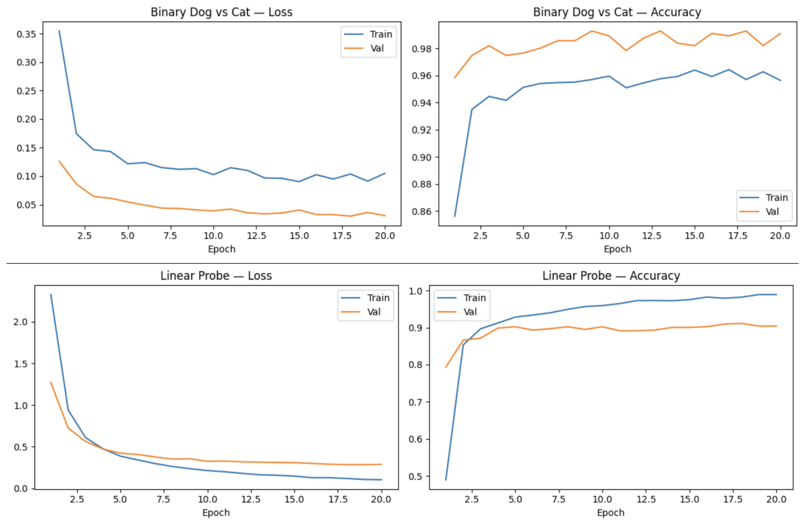

Figure 1: Binary classification and 37 classification linear probing

Unfrozen layers	Test accuracy
1	88.4%
2	88.66%
3	89.04%
4	88.06%

Table 1: Accuracies of models trained with different amounts of unfrozen layers

During training the train accuracy quickly shot up to 100%. This indicates that the model overfit to our data, which also explains why the accuracies got worse during fine-tuning as opposed to just linear probing. When additional layers are unfrozen, the model adapts more strongly to the training set, which can distort previously well-generalized pre-trained features. As a result, even though the model fits the training data better, its performance on the test set can slightly degrade due to reduced generalization.

Throughout this project, the number of epochs for training are consistently set to 20. For the implementation of gradual unfreeze, this means that we must introduce thresholds during these 20 epoch that indicate when to unfreeze the next layer. The schedule used in this project is the following:

```
unfreeze_schedule = {5:model.layer4, 10:model.layer3, 15:model.layer2}
```

This method gives a more nuanced way of training, allowing later layers to be entirely retrained towards our new classification problem while simply nudging the earlier layers that handle feature extraction. In this project, gradual unfreeze was able to attain 90.24% accuracy on the test set.

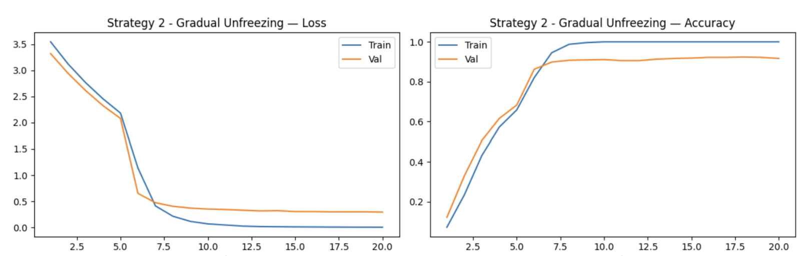

Figure 2: Gradual unfreeze cost and accuracy plots

5.2 Less data

As previously mentioned, the main problem with training neural networks is that they require large amounts of training data to get good performance. This project therefore experimented how transfer learning holds up towards when presented with limited data. Both unfreezing three layers simultaneously and gradual unfreeze were explored.

Percent of the data	100 %	10%	1%
l = 3	89.04%	65.25%	20.63%
Gradual unfreeze	90.24%	71.21%	28.92%

Table 2: Accuracies of our transfer learning methods when trained on different amounts of data

From this we can gather that performance does suffer significantly when decreasing the data, but still remains relatively high. The gradual unfreeze approach seems to also have increased importance as we introduce less data.

5.3 Regularization

To combat the apparent overfitting problem, the project introduced data augmentation as one possible solution. Regularization becomes increasingly important the less data you have, so the project also looked at how data augmentation affected the performance on models trained on a percentage of the data as before. The augmentation introduced in this project were random zoom in crops, random horizontal flips and random rotations of up to 15 degrees.

Percent of the data	100 %	10%	1%
l = 3	89.94%	69.07%	19.13%
Gradual unfreeze	87.76%	73.56%	31.37%

Table 3: Accuracies of our transfer learning methods when trained on different amounts and augmented data

These results show a slight decrease for 100% of the data, and a slight increase for 10% and 1% of the data. If we were to have a more extensive test set, the accuracy for 100% would probably have been higher as well.

Another method of regularization the project experimented with was L2-regularization that punishes too extremely fitted networks and promotes overall stability. This project used an L2 constant = 1e-4. This could be improved in future projects by doing a parameter search. Gradual unfreeze could also benefit from having an adaptive learning rate which was not implemented here.

Percent of the data	100 %	10%	1%
l = 3	89.13%	68.06%	19.62%
Gradual unfreeze	72.94%	72.94%	29.22%

Table 4: Accuracies of our transfer learning methods when trained on different amounts of the data with L2-regularization

We can see that L2 regularization had a similar effect on the performance as data augmentation apart from actually having a significantly worse performance with gradual unfreeze at 100%. This is likely due to us not having implemented adaptive learning rate or doing any parameter searching.

5.4 Imbalanced classes

As previously mentioned, a lot of real world datasets have imbalanced class-distributions. This project tested what this meant for transfer learning by artificially creating an imbalance in the Pet-dataset. In the following experiment 80% of the images from all cat breeds were removed. To get a clearer image of what the model was fitting, the evaluation also included calculating per class accuracy, cat vs dog accuracy, F1-scores and confusion matrices. The per class accuracies and confusion matrices can be seen found in the appendix.

	Overall Accuracy	Overall F1 Score	Cat Accuracy	Cat F1 Score	Dog Accuracy	Dog F1 Score
l = 3	83.16%	0.8277	69.06%	0.7407	89.86%	0.8965
Gradual unfreeze	85.77%	0.8542	73.46%	0.7727	91.63%	0.9143

Table 5: Accuracies and f1-scores of our transfer learning methods when trained on imbalanced dataset

Table 5 shows that performance on the cat classes are significantly worse here than on the dog classes. This is because the network accumulates less loss by ignoring the cat classes and by just training to fit the majority class.

A common tool to salvage imbalanced datasets is by using weighted cross-entropy. Weighted cross-entropy assigns a higher loss for miss-classifying rarer classes, which makes sure the network can't just ignore them during training.

	Overall Accuracy	Overall F1 Score	Cat Accuracy	Cat F1 Score	Dog Accuracy	Dog F1 Score
l = 3	83.62%	0.8335	71.85%	0.7551	89.22%	0.8897
Gradual unfreeze	85.39%	0.8509	73.71%	0.7741	90.95%	0.9074

Table 6: Accuracies and F1-scores of our transfer learning methods when trained on imbalanced dataset and implemented weighted cross-entropy

Table 6 shows how implementing weighted cross-entropy did slightly increase the overall accuracy and F1-score for cats, while also sightly dragging down the dog accuracy as a result. Despite that, the overall accuracy remains relatively the same.

6 Conclusion

This project showed that transfer learning can be used to attain good accuracy on the Oxford-IIIT Pet-dataset, getting up to 89% accuracy with 3 unfrozen layers. It experimented with gradual unfreezing and showed how it could boost performance, especially on smaller datasets. It showed that even greater performance can be attained by applying different regularization techniques, such as data augmentation and L2 regularization. Finally the project experimented with imbalanced datasets and took a detailed look on the per class performances with and without weighted cross-entropy to mitigate the disparity.

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